

PHYS 6213 Project

I will assign your grade in this course based on a written report, 4-10 pages in length, and a short 5-minute ‘show and tell’ about your project.

Report deadline and presentations: July 8.

Here is the prompt:

Propose an application of machine learning tools in your research area. Choose something where you already have some domain expertise and have some confidence in assessing outputs. For example, this could be a computation you’ve already done, a data analysis you’ve created yourself but could be made more efficient, etc. Explain in detail the research problem and why machine learning tools could be useful or convenient. Use an LLM model as a research assistant to help you implement your idea. Describe in detail how you used the LLMs – did you use agents? Did seek assistance with deriving equations? Learning new concepts? Manipulating data? Writing code? Describe in detail how you validated the outputs; what did you find was accurate and inaccurate? If you used a neural network, describe why you chose a particular architecture. If you used LLMs, describe how you engineered your prompts or agents and justify these choices. For this project, you are welcome to use any tool in any way you see fit. The only thing I ask is the following: since I as a human will be reading your reports, I would like for you as a human to write the reports. A reasonable length for your report is 4-5 pages. Please keep the length under 10 pages. For the show-and-tell presentation, give a short ‘pitch’ for your project describing the problem and why machine learning is useful, tell us the outcome, and then tell us a few things about what you found LLMs useful for.

Assessment: Pass-Fail – if you complete the write-up and presentation as instructed, you will receive full marks.